Year 10 Mathematics

Number 1

1.2 Order of Operations (BIDMAS)

Order of Operations (BIDMAS)

STARTER · 5 MIN

Date: _____ **Title:** Order of Operations (BIDMAS)

LEARNING OBJECTIVES

- Understand and use the correct order of operations (BIDMAS)
- Work out calculations with brackets and a fraction line
- Use brackets correctly on a calculator

STARTER *Answer in your book. No calculators.*

1. Work out 5^2

2. Work out $\sqrt{49}$

3. Work out 2^3

4. Work out $3 + 4 \times 2$

Starter — Answers

STARTER

Tick or correct in green pen

1. $5 \times 5 = 25$

2. $7 \times 7 = 49$, so 7

3. $2 \times 2 \times 2 = 8$

4. $3 + 8 = 11$ (not 14)

Success Criteria

SUCCESS CRITERIA

By the end of this lesson:

BRONZE	I can understand and use the correct order of operations (BIDMAS)	☐
SILVER	I can work out calculations with brackets and a fraction line	☐
GOLD	I can use brackets correctly on a calculator	☐

Tick each box in the plenary when you can do it on your own.

Introduction — BIDMAS

So that everyone gets the same answer, we always work in the same order.

1st B	2nd I	3rd D M	4th A S
Brackets $(5 - 2) = 3$	Indices (powers and roots) $2^2 = 4$	Divide and multiply, from left to right $12 \div 3 \times 2 = 8$	Add and subtract, from left to right $10 - 4 + 3 = 9$

Examples 1 and 2 — Using BIDMAS

EXAMPLES · 5 MIN

BRONZE

Watch and copy into your book

Example 1

Work out $7 + 3 \times 5$

Example 2

Work out $18 - 12 \div 3 + 2^2$

Examples 1 and 2 — Corrections

CORRECTIONS

BRONZE

Check your copy and correct in green pen

Example 1

$7 + 3 \times 5$

$= 7 + 15$

$= \mathbf{22}$

Tip: multiply before you add.

Example 2

$18 - 12 \div 3 + 2^2$

$= 18 - 12 \div 3 + 4$

$= 18 - 4 + 4$

$= \mathbf{18}$

Tip: add and subtract from left to right.

Task 1

PRACTICE · 9 MIN

BRONZE

Show each step on a new line

1. Work out $12 + 4 \times 2$

2. Work out $11 - 3^2$

3. Work out $20 \div 4 + 6 \times 3$

4. Work out $12 - 2^2 \times 3$

5. Work out $5 + 10 \div 2 - 3$

6. Work out $3 \times 4^2 - 10$

Task 1 — Answers

CORRECTIONS

BRONZE

Tick or correct in green pen

1. $12 + 8 = 20$
2. $11 - 9 = 2$
3. $5 + 18 = 23$
4. $12 - 12 = 0$
5. $5 + 5 - 3 = 7$
6. $3 \times 16 - 10 = 38$

Examples 3 and 4 — Brackets and the Fraction Line

EXAMPLES · 5 MIN

SILVER

Watch and copy into your book

Example 3

Work out

$7 - 3 \div (5 - 2) \times 2^2 + 5$

Example 4

Work out $\dfrac{3 \times 4 + 2^2}{10 - 2 \times 3}$

Examples 3 and 4 — Corrections

CORRECTIONS

SILVER

Check your copy and correct in green pen

Example 3

$7 - 3 \div (5 - 2) \times 2^2 + 5$

$= 7 - 3 \div 3 \times 2^2 + 5$

$= 7 - 3 \div 3 \times 4 + 5$

$= 7 - 1 \times 4 + 5$

$= \mathbf{8}$

Tip: one step per line.

Example 4

Top: $3 \times 4 + 2^2 = 12 + 4 = 16$

Bottom: $10 - 2 \times 3 = 10 - 6 = 4$

$\frac{16}{4} = \mathbf{4}$

Tip: the fraction line acts as a bracket.

Task 2

PRACTICE · 9 MIN

SILVER

Show each step on a new line

1. Work out $(12 + 4) \times 2$

2. Work out $(8 - 3 \times 2)^2$

3. Work out $5 + (5 \times 2)^2 \div 5$

4. Work out $\dfrac{4 + 4^2}{6 \div 3 \times 2}$

5. Work out $\dfrac{16 - 4 \times 3}{6 \div 3 \times 2}$

6. Work out
$3 + 2 \div (7 - 9) \times (5 \times 2 - 6)$

Task 2 — Answers

CORRECTIONS

SILVER

Tick or correct in green pen

1. $16 \times 2 = 32$

2. $2^2 = 4$

3. $5 + 100 \div 5 = 25$

4. $\frac{20}{4} = 5$

5. $\frac{4}{4} = 1$

6. $3 + 2 \div (-2) \times 4 = 3 - 4 = -1$

Examples 5 and 6 — Brackets in Problems

EXAMPLES · 5 MIN

GOLD

Exam-style: show all your working

Example 5 (1 mark)

Insert one pair of brackets to make this correct:

$4 + 6 \times 5 - 2 = 48$

Example 6 (2 marks)

Use your calculator to work out

$$\frac{6.2^2 - 3.1}{\sqrt{4.8} + 2.6}$$

Give your answer correct to 3 significant figures.

Examples 5 and 6 — Corrections

CORRECTIONS

GOLD

Check your copy and correct in green pen

Example 5

$(4 + 6) \times 5 - 2$

$= 10 \times 5 - 2$

$= 50 - 2$

$= 48$ ✓

Tip: try each place and check.

Example 6

$(6.2^2 - 3.1) \div (\sqrt{4.8} + 2.6)$

$= 7.376499\ldots$

$= \mathbf{7.38}$ (3 s.f.)

Tip: write the full display, then round.

Task 3

PRACTICE · 9 MIN

GOLD

Exam-style: show all your working

1 Insert brackets to make this correct:

$4 \times 5 - 3 + 2 = 10$

(1 mark)

2 Insert brackets to make this correct:

$2^2 + 5 \times 2 + 3^2 = 99$

(1 mark)

3 Use your calculator to work out $\dfrac{24.3 - 16.8}{0.18} + \sqrt{67.4}$

Write down all the figures on your calculator display.

(2 marks)

4 Use your calculator to work out $\dfrac{(3.6 + 2.1)^2}{5.4 - 1.85}$

Give your answer correct to 3 significant figures.

(2 marks)

Task 3 — Answers

CORRECTIONS

GOLD *Tick or correct in green pen*

1 $4 \times (5 - 3) + 2 = 10$

2 $(2^2 + 5) \times (2 + 3^2) = 9 \times 11 = 99$

3 $41.666\ldots + 8.2097\ldots = 49.87641697$

4 $\frac{32.49}{3.55} = 9.152112\ldots = 9.15$ (3 s.f.)

Plenary — Exit Ticket

PLENARY · 5 MIN

Answer in your book. Try all three.

BRONZE Work out $6 + 12 \div 3$

SILVER Work out $(2 + 3)^2 - 4 \times 5$

GOLD Work out $\dfrac{5^2 - 1}{3 \times 4 - 4}$

Reflect: Look back at the learning objectives. Which ones can you do now?

Plenary — Answers

PLENARY

Tick or correct in green pen

BRONZE $6 + 4 = 10$

SILVER $25 - 20 = 5$

GOLD $\frac{24}{8} = 3$

Exam Practice — Using BIDMAS

EXAM PRACTICE · 10 MIN

BRONZE

IGCSE exam-style: answer on the line, show all your working

Question 1

(2 marks)

(a) Work out $20 - 3 \times 4$ (1)

.....

(b) Work out $(2 + 3)^2 - 4^2$ (1)

.....

Question 2

(2 marks)

Ali says $6 + 4 \times 3 = 30$

(a) Explain the mistake Ali has made. (1)

.....

(b) Work out the correct answer. (1)

.....

Exam Practice — Brackets and Fraction Lines

EXAM PRACTICE · 10 MIN

SILVER

IGCSE exam-style: answer on the line, show all your working

Question 3 (2 marks)

Work out

$$\frac{5 + 3^2}{9 - 2}$$

.....

Question 4 (2 marks)

Insert one pair of brackets to make each statement correct.

(a) $3 + 5 \times 4 - 1 = 31$ (1)

.....

(b) $12 \div 2 + 4 \times 3 = 6$ (1)

.....

Exam Practice — Calculator Skills

EXAM PRACTICE · 10 MIN

GOLD

IGCSE exam-style: answer on the line, show all your working

Question 5

(2 marks)

Use your calculator to work out

$$\frac{\sqrt{19.6 + 3.2^2}}{2.1 \times 0.4}$$

(a) Write down all the figures on your calculator display. (1)

.....

(b) Give your answer correct to 3 significant figures. (1)

.....

Question 6

(2 marks)

The distance, s metres, a car travels while braking is

$$s = \frac{v^2 - u^2}{2a}$$

Work out s when v = 12.5, u = 3.8 and a = 2.4. Give your answer correct to 3 significant figures. (2)

..... m

Exam Practice — Mark Scheme

CORRECTIONS

BRONZE

Mark your work: M = method mark, A = accuracy mark, B = independent mark

Question 1

(a) $3 \times 4 = 12$, $20 - 12$

= 8 B1

(b) $5^2 - 4^2 = 25 - 16$

= 9 B1

Tip: brackets first, then powers.

Question 2

(a) He added before multiplying B1

(should do 4×3 first)

(b) 6 + 12 = 18 B1

Tip: × and ÷ come before + and −.

Exam Practice — Mark Scheme

CORRECTIONS

SILVER

Mark your work: M = method mark, A = accuracy mark, B = independent mark

Question 3

top: 5 + 9 = 14, bottom: 7 — M1

14 ÷ 7 = 2 — A1

Tip: a fraction line acts like brackets.

Question 4

(a) (3 + 5) × 4 − 1 = 31 — B1

(b) 12 ÷ (2 + 4) × 3 = 6 — B1

Tip: test each position, then check.

Exam Practice — Mark Scheme

CORRECTIONS

GOLD

Mark your work: M = method mark, A = accuracy mark, B = independent mark

Question 5

19.6 + 10.24 = 29.84

(a) 6.502… (6.50309…) B1

(b) 6.50 B1

Tip: put brackets round the top and bottom.

Question 6

$12.5^2 - 3.8^2 = 141.81$ M1

141.81 ÷ 4.8 = 29.54…

s = 29.5 m A1

Tip: type $(12.5^2 - 3.8^2) \div (2 \times 2.4)$.